Track zero sensor for Commodore 1541-II

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In this document, a track zero[1] sensor for Chinon drives used in the Commodore 1541-II family of floppy drives is presented. The purpose of the sensor is to get rid of the notorious bumping these drives exhibit when trying to establish the outermost track position of the head, for instance when starting to format a disk. Adding the functionality involves expanding the mechanism with the sensor, modifying the main PCB in the drive and changing the ROM software to support it.

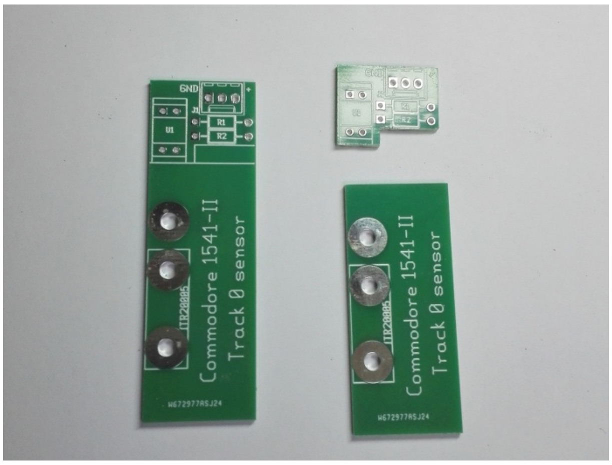
Identifying the mechanism model

The Chinon mechanism was used in the 1541-II, but not in the 1541 and features a turn handle to lock the floppy. To properly identify it, you have to open the drive, and look for the identification Chinon on the mechanism or compare the mechanism with the photographs in this document. The 1541-II was produced with at least three different mechanisms, the one supplied by Chinon is believed to be the most common.

Building the sensor (Chinon)

To build the sensor part, follow these steps:

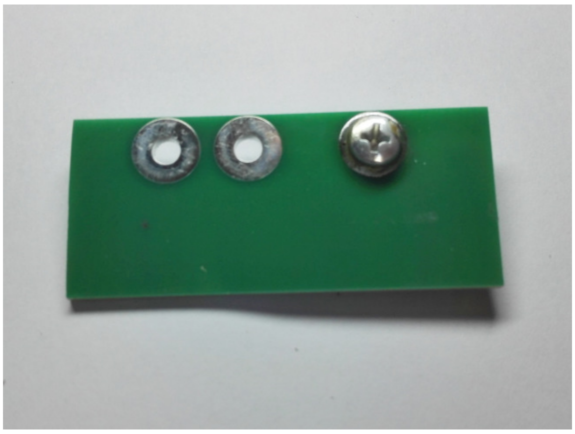
Cut the PCB in two parts as shown using a saw or Dremel tool. Also remove the bottom part of the smaller PCB indicated by the silk lines using a file, Dremel tool and/or saw:



[1] To be ultra correct: Commodore DOS as present in the 1541-II does not have a track zero. The outermost track in this DOS has the number one. However, the device dealt with in this document is generally known as the track zero sensor. For this reason, the term was not changed although in this context it detects the condition "head at outermost track".

For the next step you need two M3x6 bolts, an M3 nut and the base plate PCB. The nut and bolts need to be solderable; stainless steel ones are not. The author used zinc plated hardware, brass should also work.

Insert a bolt through the hole for the optical sensor closest to the board edge, with the head of the bolt on the side without text. Secure the bolt with the nut. Solder the head of the bolt to the tin plated area surrounding it.



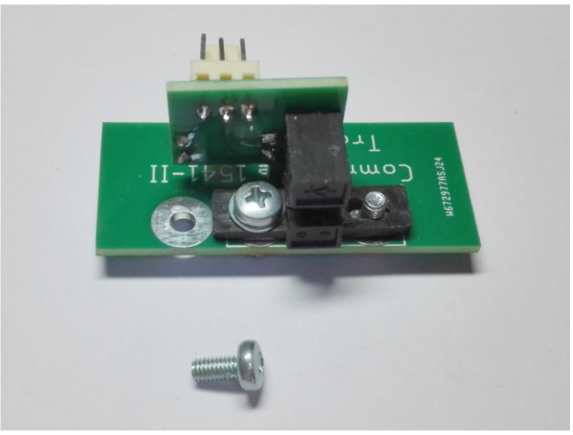
The protruding threaded part of the bolt will serve as a guide pin for the photo sensor.

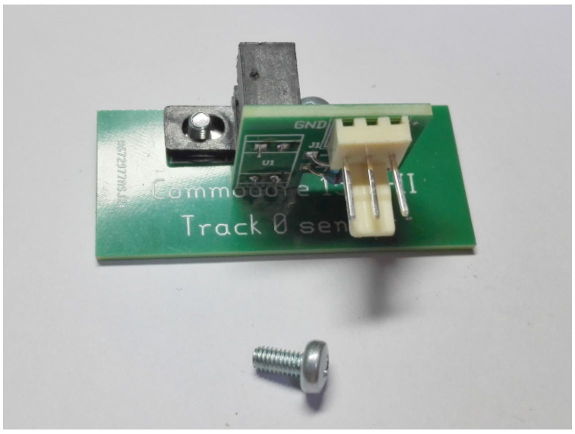
Remove the nut (after the lot has cooled down). Insert a second M3x6 bolt in the other hole for the optical sensor, this time the head is on the text side of the base plate. Secure the bolt with the nut and solder the nut to the tin plated area surrounding it.

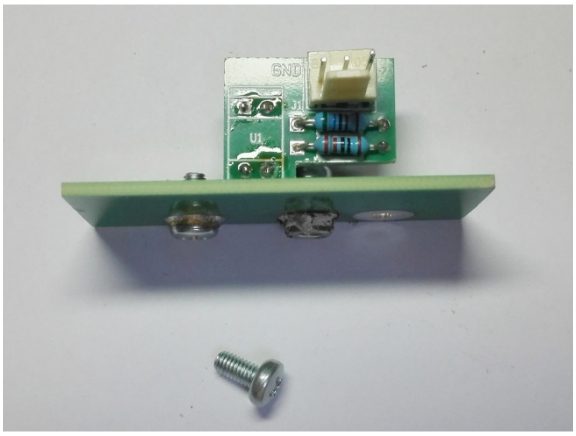
Remove the bolt (after cooling down). Preparation for the base plate is now complete.

Build up the small PCB. See the BOM (appendix A) for the part details. The KF2510 connector shown is optional, but recommended. The photo sensor is mounted on the non-text side of the PCB and soldered on the text side. The two resistors and the connector are located on the other side and soldered on the non-text side (see photos below). Attach the assembled smaller PCB to the base plate using an M3x6 bolt and an M3 small washer.

The whole thing should look like this:







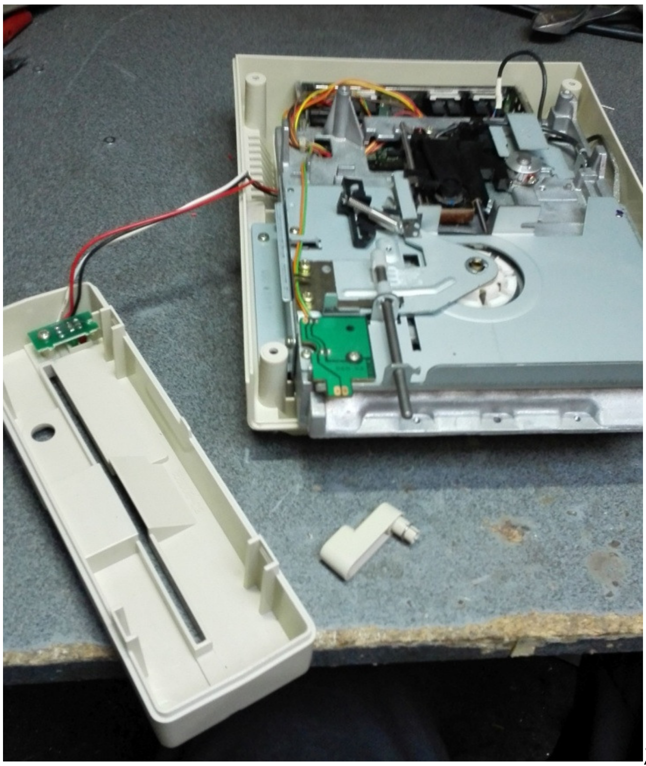
As can be seen, the thread part of the first bolt soldered is used as a guide pin and the sensor can be shifted sideways by loosening the second bolt, setting its position and tightening the second bolt again. The remaining hole in the base plate and the third bolt shown are used to mount the sensor assembly in the floppy mechanism.

After checking everything fits and slides smoothly, remove the smaller PCB from the base plate.

Mounting the sensor in the drive (Chinon)

Take the 1541-II apart in the following order:

- Remove the four self tapping screws that attach the top cover to the bottom of the case;


- Remove the top cover and lay it aside;
- Remove the disk locking handle by pulling. Lay the handle aside;
- The front can now be taken off by unhooking the tabs at the bottom. Although not strictly required, the front can be separated from the rest by removing the self tapping screw that attaches the PCB holding the LEDs to the front. Move the front away or lay it aside. You should now have this:

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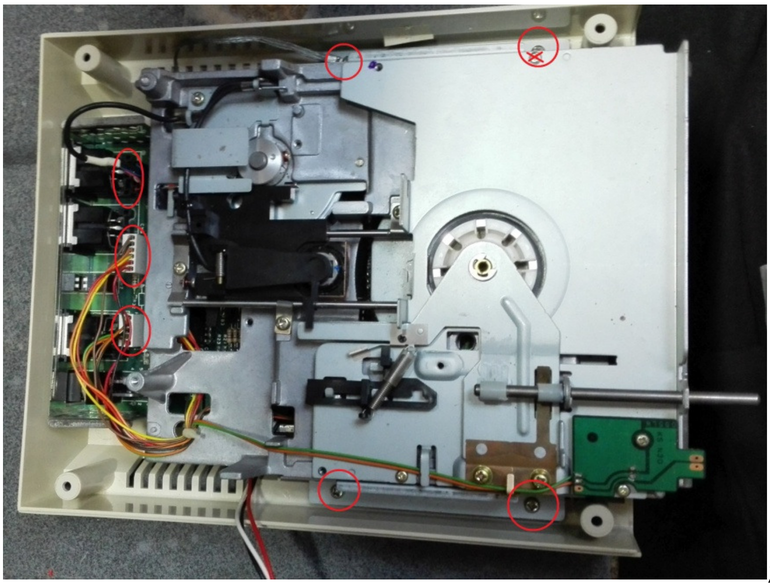
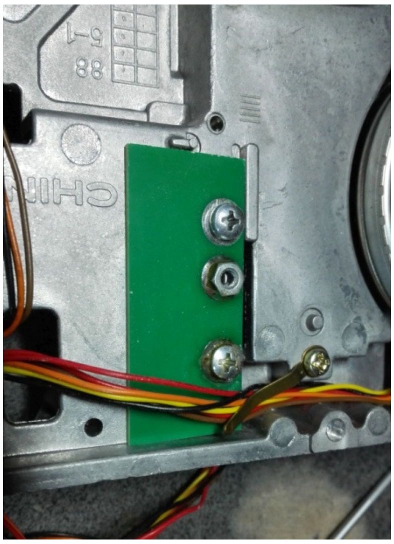
- The drive mechanism is connected to the main PCB with three connectors at the rear. Remove the three connectors from the main PCB by carefully pulling them off noting the orientation of the connectors. If you are not sure to remember the orientations, take a detailed photograph. The connectors involved are indicated with red oval circles in the photograph below:



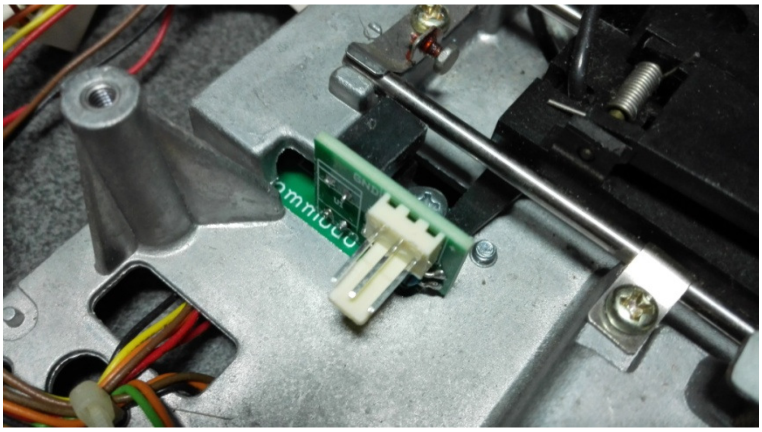
;

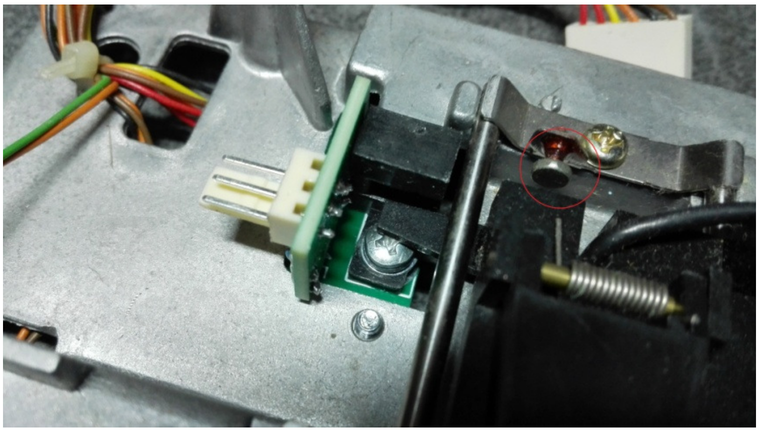
- The drive mechanism is mounted to the bottom part of the case with four self tapping screws indicated with red circles in the photograph above. Loosen these and lay them aside. Note that the screw at the right rear also connects a ground strap. The drive mechanism can now be taken out;
- For the moment lay the bottom case with the main PCB aside;
- Put the drive mechanism upside down with the rear facing you. Mount the base plate of the sensor using the third M3x6 bolt, using the tapped hole already present in the mechanism's chassis:



- Turn the drive mechanism over. Carefully move the head assembly all the way towards the spindle. The top side will look like this:

Mount the sensor in place. First hook the rear hole in the photo sensor over the rear bolt on the base plate and then attach the optical sensor using an M3x6 bolt and the small washer. The end result should look like this:





Carefully move the head assembly to the rear of the mechanism and check that the vane on the head assembly does not touch the photo sensor. Note: the bolt circled in the photograph is the set screw mentioned in the alignment procedure;

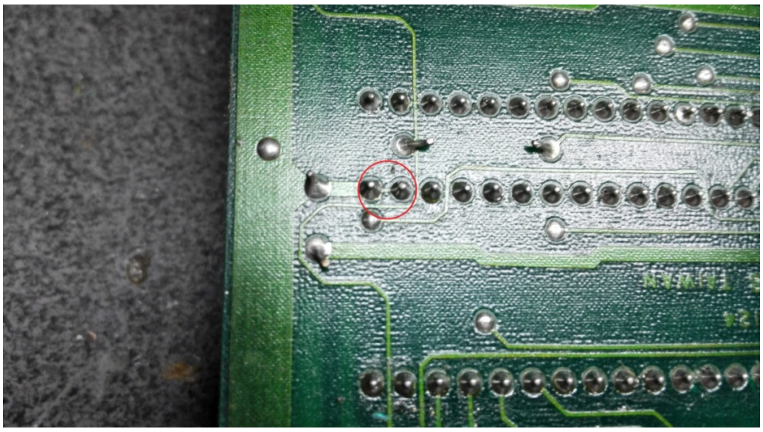
- The drive mechanism is now ready for wiring and adjustment.

Changing the main PCB (1541-II, Chinon)

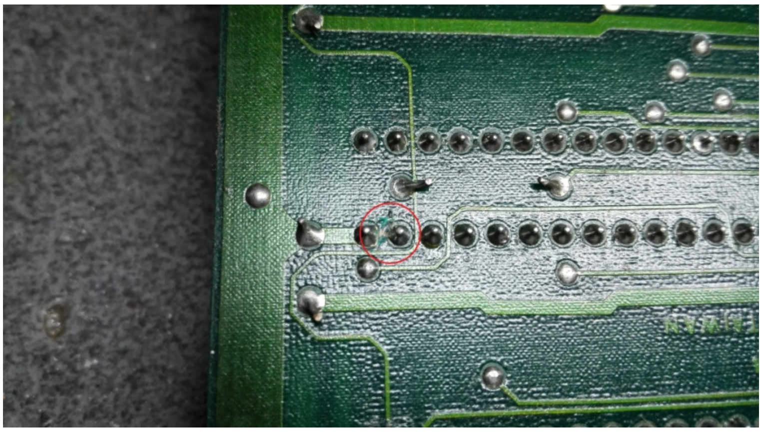
- The main PCB is in the bottom part of the case:

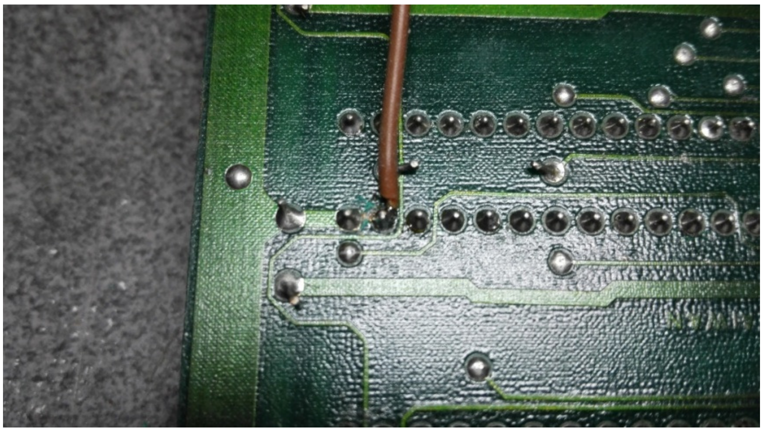
  Take the PBC out after removing the three self tapping screws holding it. There is one screw at the front and two at the back. The screw at the right rear holds the other end of the ground strap mentioned before;
- Turn the PCB over (solder side up) and locate pin 1 of the front 6522 VIA. It is marked as a square island and has a wide ground copper trace running from it, the other pins have round solder islands. Pin 1 of the 6522 VIA is connected to pin 2 with a small copper trace:

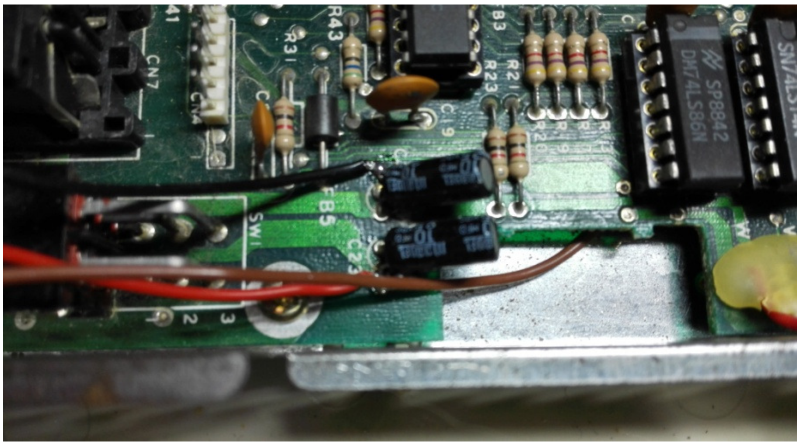
- Cut the trace between pins 1 and 2 with a sharp object. Take care not to damage the solder islands. Check for no continuity between pins 1 and 2. The result should look like this:

- Solder a wire to pin 2 of the VIA. In the following text the wire will be referred to and shown as the brown wire. It carries the track 0 signal and will be connected to the middle connector pin of the sensor assembly:



- Mount the PCB back in the bottom part of the cabinet. Lead the brown wire through the opening on the left side of the PCB. Use the three middle length self tapping screws and do not forget the ground strap at the right rear;
- On the top of the main PCB toward the rear, solder two wires to the connections of C23 and C22 as shown in the photograph below. The wire on C23 closest to the board edge is the power supply for the sensor and will be called and shown as the red wire in the remainder of this text. The other wire (called/shown as the black wire in the remainder of this text) is connected to the minus side of C22 (note: the minus side of C23 can also be used; but the connection on C22 is easier to work with). Lead the three wires to the rear of the case for the moment. The result should look like this:



Changing the firmware (1541-II, Chinon)

To change the firmware, proceed as follows:

- The firmware is stored in a ROM chip, located on the left front of the PCB (marked U4). On regular production 1541-II's it is the only chip in a socket. The ROM should carry the marking 251968-03[2]. Carefully remove the ROM from its socket using a small screwdriver or similar tool. This may take more effort than expected as the chip has been sitting in the socket for more than 20 years;

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[2] A small number of early production 1541-II's may have a ROM with identification 251968-02. In that case, you can leave it in place. This fact is easy recognizable: the drive will bump each time it is powered on.

- Program a 27128(A) EPROM with the last firmware version for the 1541B/C drive. Its identification is 251968-02(.bin). Insert the programmed EPROM in the vacated ROM socket taking care not to bend any pins under it. The direction of the notch on the housing of the EPROM should match the notch on the socket (and the other notched chips on the main PCB);
- The main PCB is now ready.

Assembling the whole lot (1541-II, Chinon)

Put everything together again, but for the top cover of the cabinet, as the sensor will need adjustment. Proceed as follows:

- Using the four shortest self tapping screws mount the mechanism back into the cabinet. The screw on the right rear also holds the ground strap. The three wires mounted before should be over the main PCB and under the mechanism towards the rear;
- Reconnect the three connectors emanating from the drive to the matching connectors on the main PCB. If you took photographs before, check if they are mounted in the correct orientation. The black head connector only fits in one way. The others cannot be interchanged easily because of the difference in pin counts and the wire lengths;
- If you want to, twist the three new wires together. Then lead them towards the track 0 sensor on the mechanism. At your option, you can use the KF2510 connectors as shown in the photographs, or directly solder the wires on the sensor. The use of a connector is recommended, because the drive mechanism can then be removed from the cabinet/main PCB more easily without soldering. Cut the wires to length and strip the ends;
- On the sensor assembly the brown wire is the centre wire. The red wire goes to the side of the connector marked + (pin 3 of the connector); the black wire is marked GND on the sensor (pin 1 of the connector). Either assemble the female KF2510 connector and put it on the receptacle or solder the wire ends in the correct positions on the sensor assembly;
- The wiring is now complete. If you want, you can check if the red and black wires were not inadvertently crossed: there should be continuity between the black wire and the ground strap and/or the mechanism metal parts;
- If previously removed, attach the LED PCB back to the front panel. Install the front panel on the hooks of the bottom of the cabinet and push the release lever back on. The drive is now ready for alignment of the track 0 sensor.

Aligning the sensor (1541-II, Chinon)

The next and final step is to align the sensor. Do the following:

- Loosen the screw holding the photo sensor to the base plate and slide the photo sensor all the way toward the rear and then fasten the screw up to the point that the photo sensor can slide, but with some effort required;
- Connect the power supply. Switch the drive on. Nothing untoward should happen. A new phenomenon will be apparent though: the mechanism should bump and a rattle similar to the one heard before when formatting a disk should occur;
- The head assembly has now retracted to the outermost track position. Take note of the distance between the set screw and the back of the head assembly: the latter should just clear the set screw with a distance of about 0.35 – 0.55 mm (note: the setscrew is a M2.5 bolt locked in place with some lacquer. Do NOT adjust it as it determines the basic track 0 alignment.);

- Switch the drive off and slide the photo sensor toward the spindle of the mechanism for about 0.3 mm [3]. Carefully move the head assembly away from the outermost track towards the spindle by hand. Switch the drive back on;
- The head again should go completely to the back. When it has, again check the clearance between the setscrew and the back of the head assembly. By repeating this and the previous point, you should be able to find a position of the photo sensor where the following two conditions are met reliably and simultaneously:
    - The head no longer rattles;
    - The distance between the setscrew and the rear of the head assembly is always the same as the one observed the first time (approximately 0.35 - 0.55 mm);
- If you reach a point where the distance suddenly is about 0.53 mm (1 track) larger, the sensor is too far inward toward the spindle. Move it back at least 0.25 - 0.3 mm and retest;
- In case to you want to be ultra precise, try and find the positions where the head just starts rattling (sensor towards the rear) and where it just stops one track too far inwards (sensor towards the front). These positions should be about 0.53 mm apart. Then slide the photo sensor to a position exactly halfway between these two points;
- Tighten the screw holding the photo sensor to the base plate, checking that the photo sensor does not move in the process;
- Power the drive off, gently move the head assembly a bit inwards and switch back on. Do this several times for a final test: the drive should move the head to the outermost track without rattling and reproduce the initial assembly / setscrew distance observed reliably. As an additional test you can leave the head in position at least once between switching off and on. In this case, the head should move one track back and forth ending again with the same set screw distance. If all that goes well, remount the top cover of the cabinet. Job done. Enjoy your rattle free drive.

If in the future the drive starts rattling on power on, move the photo sensor slightly (0.2 - 0.3 mm) towards the spindle. If disks formatted in this drive give read problems in other drives, move the photo sensor slightly (0.2-0.3 mm) to the back. Normally, no readjustment should be needed.

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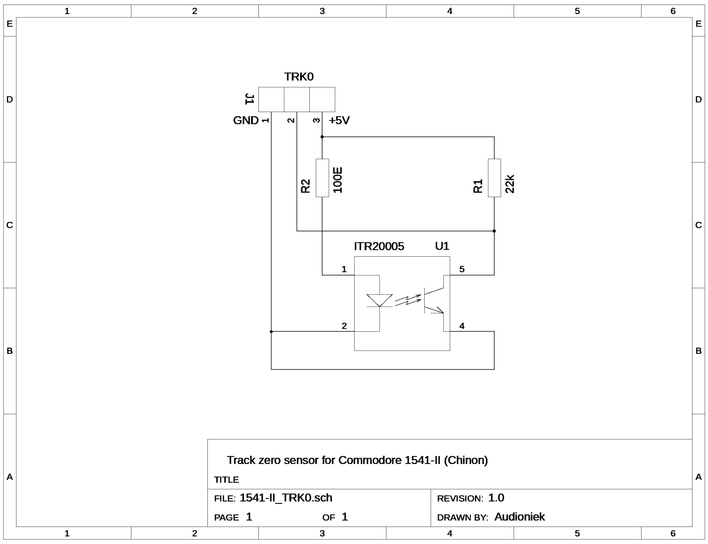
[3] To put things in perspective: the track to track distance in the 1541-II is 25.4 / 48 = 0.529 mm as it is a 48 TPI drive. Ideally the goal is to achieve a sensor position halfway between tracks zero and one.

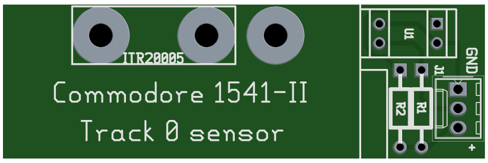
Appendix A: bill of materials

Ref.	Description	Value	Qty.	Remark
R1	Resistor, 1/4W, 0207	100E	1	
R2	Resistor, 1/4W, 0207	22k	1	
U1	Photo interrupter	ITR20005	1	
-	Bolt	M3x6	3	Solderable
-	Washer	M3, DIN125	1	
-	Nut	M3	1	Solderable
-	PCB			
J1	KF2510 male	3 pin, straight	1	Optional
J1a	KF2510 female	3 pin	1	Optional
-	Crimp contact, KF2510		3	Optional

Appendix B: schematic diagram



Appendix C: PCB layout



(True size)